**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 298 · Backtracking in computational solutions

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. When does a recursive call stop?

Q2. What is a search tree node?

Q3. Why may pruning reduce runtime?

Q4. Insert 12 into BST root10 with right child15 and left child5. Show the comparisons and position.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

backtracking — Explore choices and undo a failed path.

pruning — Discard an impossible branch.

search tree — Branches of possible decisions.

base case — Condition ending a recursive branch.

## New learning

Make a decision — At each position choose the current item or skip it. Track index and running total.

Reject a dead end — When the sum exceeds the target (non-negative inputs) or the list ends, undo the choice.

Backtrack — Try the other branch; success means a valid selection has been found. Exponential worst-case growth remains possible.

## Worked example

Find a subset totalling 6 from [2,3,4]. Trace the first failed path and successful branch.

1. Predict first — The first path over-shoots: 2+3+4=9, so reject that branch.

2. Explain the method — Undo 4, skip 3, then include 4 → 6.

# We Do · Practise with guidance

Find a subset totalling 5 from [1,3,4] using inclusion/exclusion.

```
Include 1 and 3 →4
Include 4 →8 reject
Undo 4; exclude 4 → dead end at 4
Backtrack: skip 3; include 4 →5 success
```

Predict first

Including 1 and 3 produces 4; adding 4 overshoots 5.

Explain the method

Undo 4, skip 3, then include 4 → 5.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Explain choice, dead end and backtrack using [2,3,4] target6. [6]

### Question 2

Write a recursive include/exclude Boolean procedure for subset sum with non-negative integers, including base cases. [8]

### Question 3

Explain why backtracking is useful for 4 queens but not guaranteed polynomial time. Give one safe-placement rule. [6]

## Exit Ticket · Answer before checking

Backtracking can require trying an earlier alternative after a dead end.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. When a base case is met.

Q2. A partial choice/state considered in the search.

Q3. It avoids exploring states proven unable to succeed.

Q4. 12>10 ⇒ go right to15; 12<15 ⇒ insert as left child of15.

### You Do pages · Independent questions

Q1. Try subset [2,3,4] [1], sum9 [1], prune as >6 for non-negative inputs [1], undo last choice [1], exclude 3 [1], choose 4 with 2 for sum6 [1].

Q2. Check total==target return true [2]; end-of-input false [1]; prune total>target [1]; include branch [2]; exclude branch [1]; combine with OR [1].

Q3. Rows/columns/diagonals constrained [2]; choose next row/column then recurse [1]; undo when no legal square [1]; worst-case combinatorial branching [1]; safe placement checks no shared column or diagonal [1].

### Exit Ticket · Model answer

True; reversing a choice lets the search explore another branch.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**



RHS COMPUTING / OCR H446 / mixed LESSON 298 · DRAFT

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Do Now Recall previous learning | I Do Read new knowledge | We Do Practise with guidance | You Do Work independently | Review & Improve Check understanding